

Presentation 1

• Links to presentation(s) and code(s) on GitHub:

- [Presentation Link](#)
 - Note that if the Power BI embedded in the slides fails, static visuals can be found at the bottom of the slide deck
- All coding was done in Power BI: see below // [Power BI Sharing Link](#)

• What did you do?

I created visuals for the Time Trends Dashboard. I began by compiling all CAR data into a CSV file using the import code provided by Prof. Krishna. I then loaded the file into Power BI for analysis. I then performed the following transformations on the dataset *within Power BI*:

I transformed the data using the *Group By* function to ensure each *Contact Session ID* corresponded to only one row—preventing double counting caused by multiple actions per session. I used the *Advanced Group By* option to create multiple aggregations that support deeper analysis and visualization. Specifically, I created three new columns:

- Call Start Time — *Operation*: Minimum of *Activity Start Timestamp*
Purpose: Shows timestamp of the first row in a session to identify when the call began.
- Starting Hour — *Operation*: Minimum of *Hour*
Purpose: Records the hour of day when the session started.
- All Rows — *Operation*: All Rows
Purpose: Retains information from each session in nested tables for detailed reference.

I also *expanded* the other columns from this one to access nested information.

I then used those columns to create the following columns:

- Count - *Operation*: Row/activity count per call with distinct activity timestamp
Code: `List.Count(List.Distinct([All Rows][Activity Start Timestamp]))`
Purpose: See how many different menus/actions a user goes through in each call
- Call Duration - *Operation*: Subtract the minimum timestamp of a call from the maximum

Code: Duration.TotalMinutes(
 List.Max([All Rows][Activity Start Timestamp]) -
 List.Min([All Rows][Activity Start Timestamp])
)

Purpose: Determine how long each call lasted in minutes

- **DayOfWeekNum - Operation:** Extract weekday from Call Start Time Column as number (Monday = 1 and Sunday = 7)

Code: Date.DayOfWeek([Call Start Time], Day.Monday) + 1

After these transformations, I created the following visualizations:

- **Type:** Bar Chart
X-axis: Starting Hour
Y-axis: Contact Session
Legend: DayofWeekNum
- **Type:** Line Graph
X-axis: Starting Hour
Y-axis: Average of Activity Count per Hour → **Code for Measure:**
 Average Count per Starting Hour =
 AVERAGEX(
 KEEPFILTERS(VALUES('combined_calls'[Starting Hour])),
 CALCULATE(AVERAGE('combined_calls'[Count])))
- **Type:** Line Graph
X-axis: Starting Hour
Y-axis: Average of Call Duration per Hour → **Code for Measure:**
 Count of Call Duration average per Starting Hour =
 AVERAGEX(
 KEEPFILTERS(VALUES('combined_calls'[Starting Hour])),
 CALCULATE(COUNTA('combined_calls'[Call Duration])))

*On each visual, you can filter by: Activity Name, Agent Name, EP Name, Flow Name, Queue Name, Termination Reason, and both axes and legend.

- **How does it help the project?**

This analysis of peak call time trends with various filters can help the team uncover operational inefficiencies and develop strategies to improve overall intake performance, as seen in my presentation. Based on my analysis, changing the messaging for times users are encouraged to call could increase the quality of service for users and allow agents to reach more people overall. Additionally, developing a system for tracking call outcome/success will enable deeper analysis of call effectiveness, help identify when and why shorter calls are less successful, and guide future staffing and training decisions.

- **Issues Resolved After Feedback:**

- Activities with the exact same timestamp were counted as one and activities with different timestamps were counted as distinct.
- All 3 plots pull from the CAR data to ensure consistency.
- Conclusions about peak call times were adjusted to accurately reflect the dashboard.
- Recommendations were expanded to include all 3 plots and made more precise/actionable.

- **Next steps**

For the next presentation, I plan to:

- *Dive deeper into the EP/Activity/Flow/Queue filters* to clearly display trends in peak call times by different groups or legal issues
 - Use heatmaps or line charts to highlight patterns across hours and days.
- *Analyze average activities per call and call durations by weekday* in addition to time of day to discover new trends in calling behavior
- *Determine a metric/indicator for call success* which I observe according to time trends

- **References (Mention if you built up on someone else's work)**

Code import files from Prof. Krishna